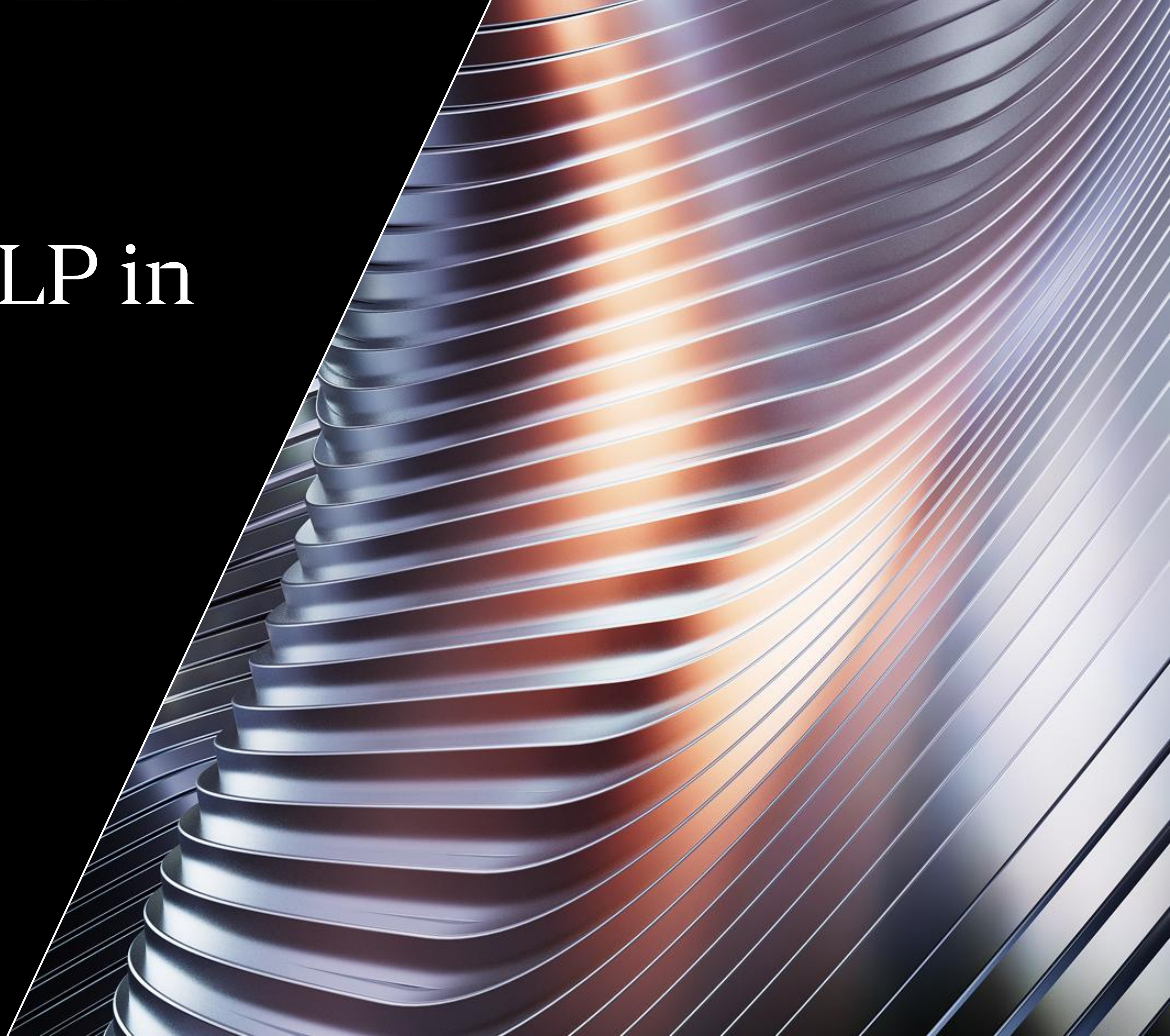
SoftServe

Free Text and NLP in Data Science

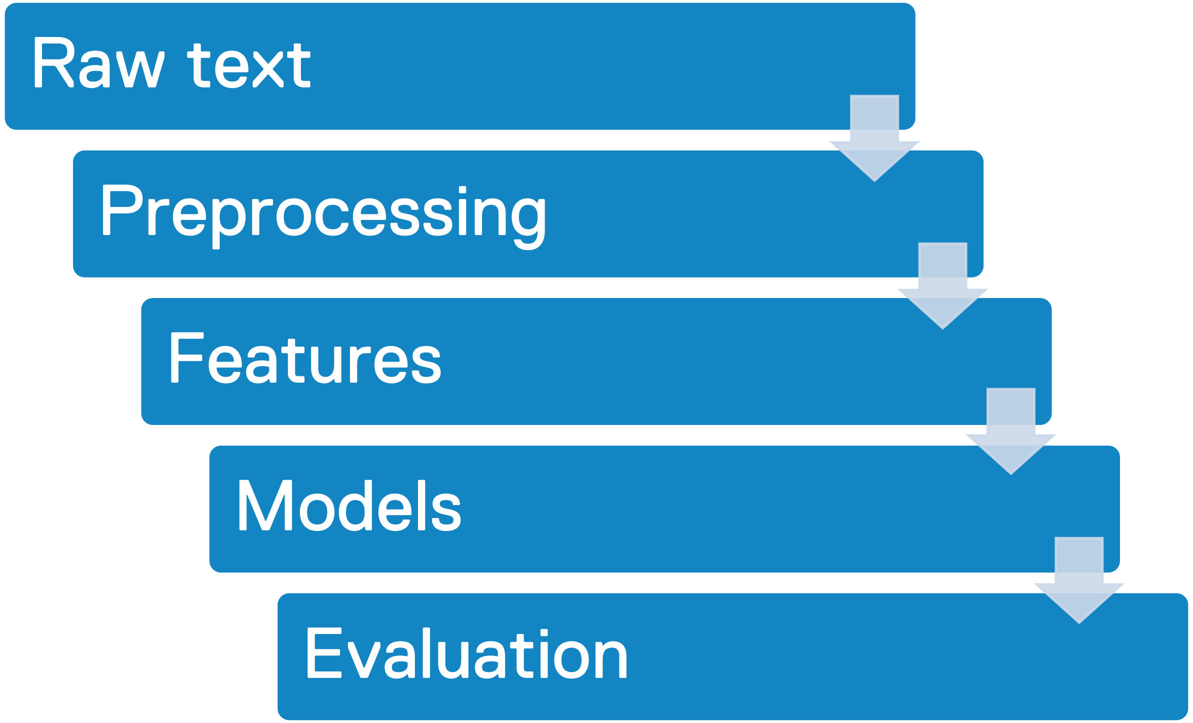
LEÓN JARAMILLO

SOFTSERVE ACADEMY MENTOR



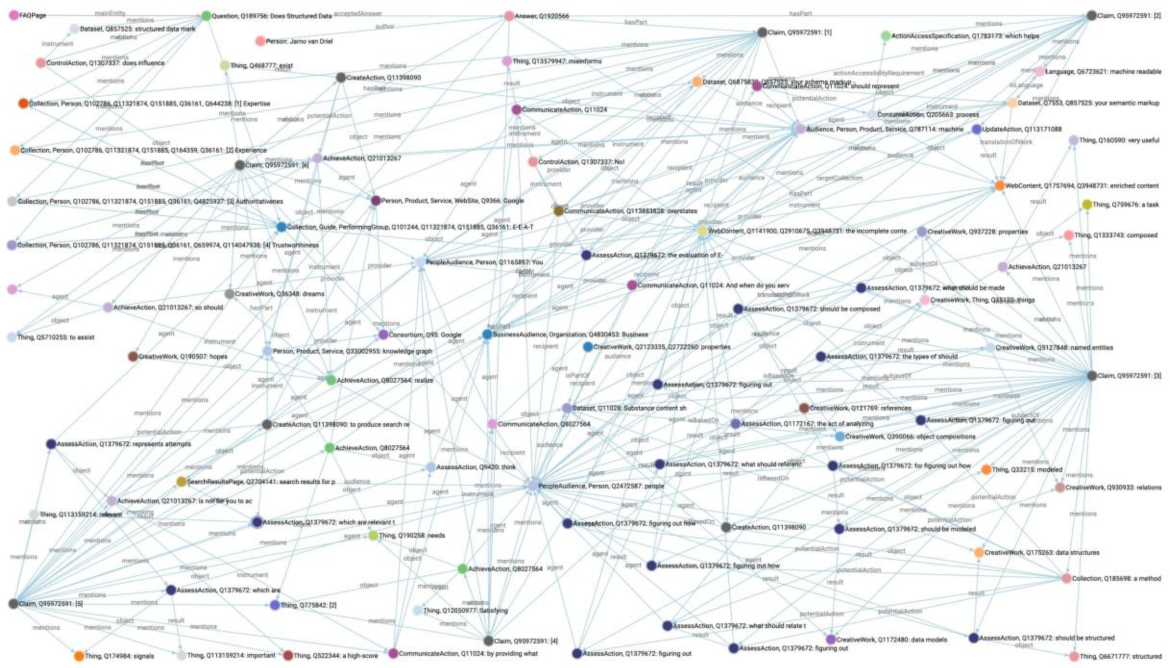
LEARNING OBJECTIVES

- Explain what free text is and why it is difficult to analyze.
- Describe core text preprocessing steps.
- Compare BoW, TF-IDF, embeddings, and n-grams.
- Explain how RNNs, Transformers, and transfer learning changed NLP.
- Describe an end-to-end NLP project pipeline.
- Choose suitable evaluation metrics for NLP tasks.



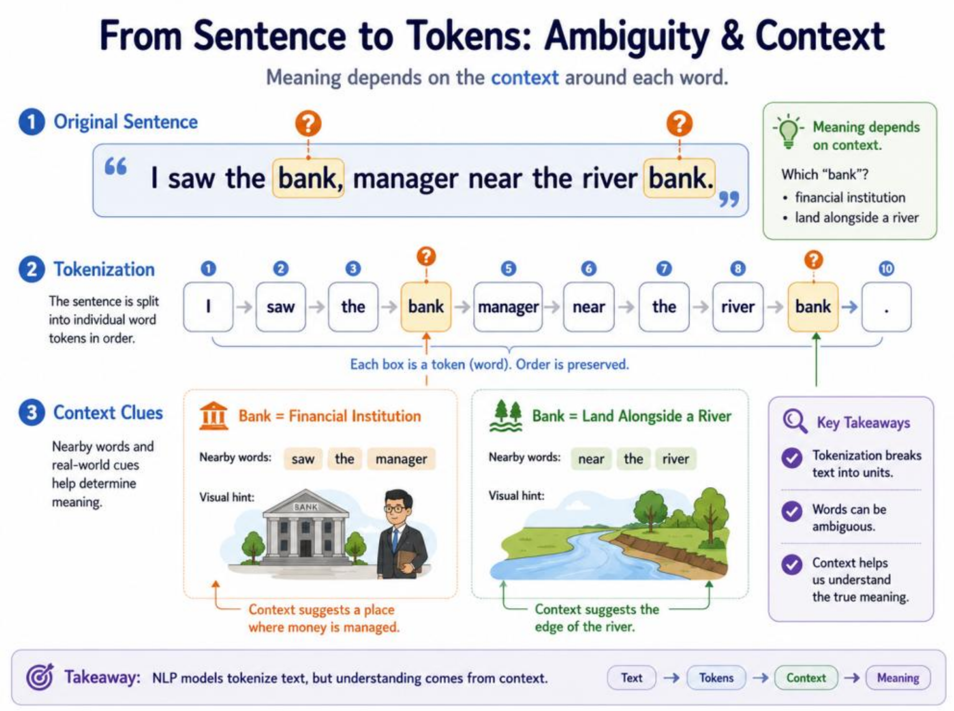
WHY FREE TEXT MATTERS

- Much real-world data is text: reviews, tickets, emails, chats, logs, documents, web pages.
- Text is rich but messy.
- Unlike tables, text does not come with obvious columns and types.
- Free text refers to human language data that has not been organized into predefined fields, categories, or rigid formats.
- NLP helps extract patterns, signals, and predictions from language.



FREE TEXT IS STRUCTURED, BUT NOT EASILY

- Humans understand context, ambiguity, tone, and references.
- Machines need explicit representations.
- **Example challenge:** pronouns, sarcasm, negation, missing context.
- NLP often starts by extracting useful signals without full human-level understanding.



KEY TERMINOLOGY

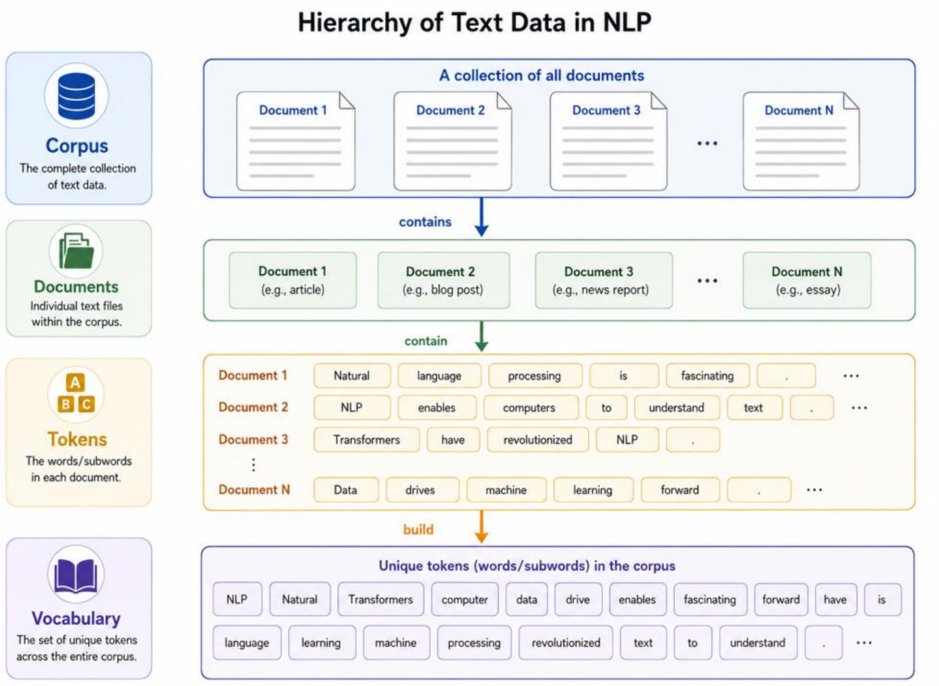
Document: one text unit, such as a review or support ticket.

Corpus: a collection of documents.

Token: a unit of text, often a word or punctuation mark.

Vocabulary: the unique tokens in the corpus.

Feature: a numeric representation usable by a model.



THE NLP PIPELINE AT A GLANCE

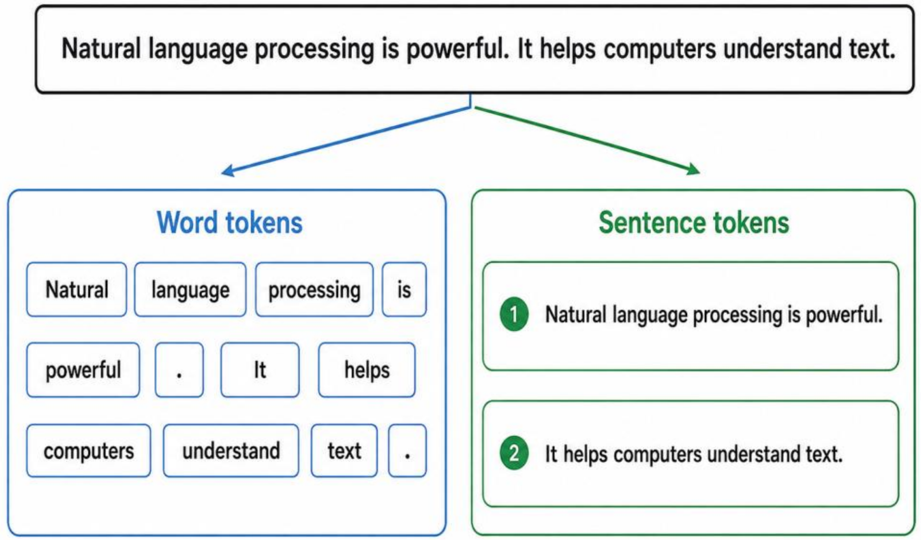
1. Collect text
2. Clean and normalize it
3. Convert text into numeric features
4. Train or apply a model
5. Evaluate results
6. Deploy and monitor

TEXT CLEANING

- Lowercasing.
- Removing punctuation, numbers, or special characters when appropriate.
- Removing stopwords.
- Handling noise from HTML, social media, OCR, logs, or forms.
- Cleaning decisions depend on the task.

TOKENIZATION

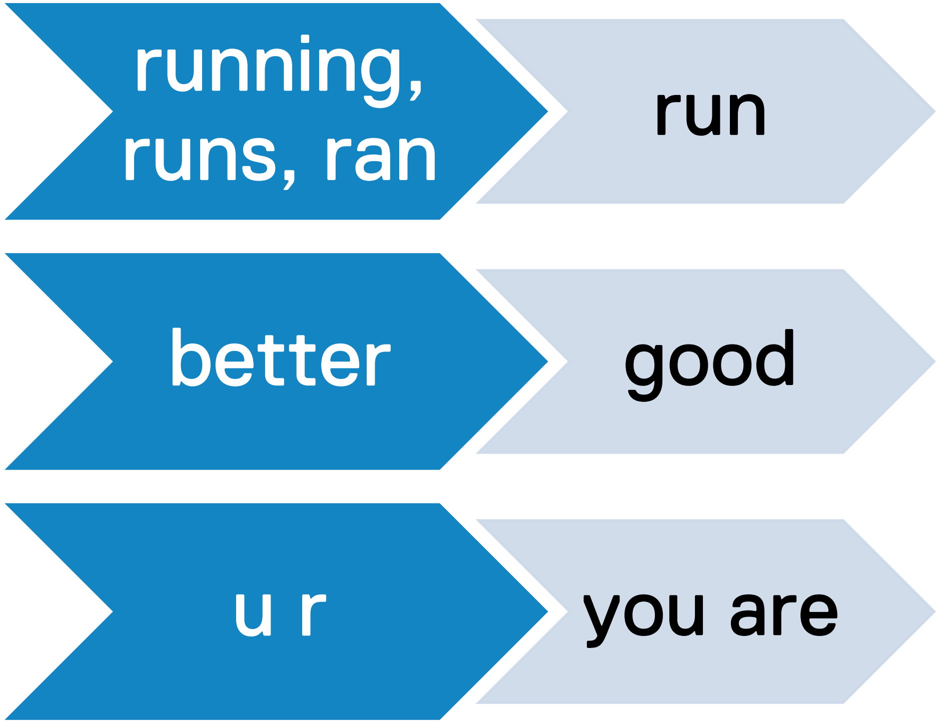
- **Tokenization** splits text into smaller units.
- Word tokenization supports frequency analysis and feature extraction.
- Sentence tokenization helps with summaries, translation, and document structure.
- Tokenization is language-dependent.



STEMMING, LEMMATIZATION, AND NORMALIZATION

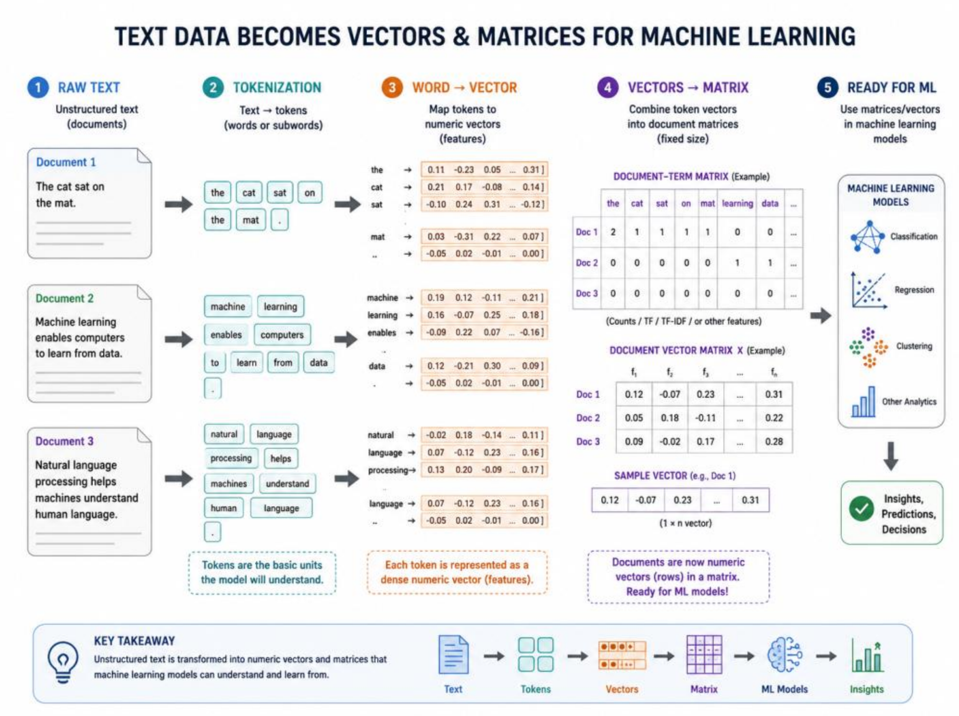
- **Stemming:** cuts words to rough roots.
- **Lemmatization:** maps words to dictionary forms.
- **Normalization:** handles spelling, abbreviations, synonyms, and casing.

Goal: reduce unnecessary variation.



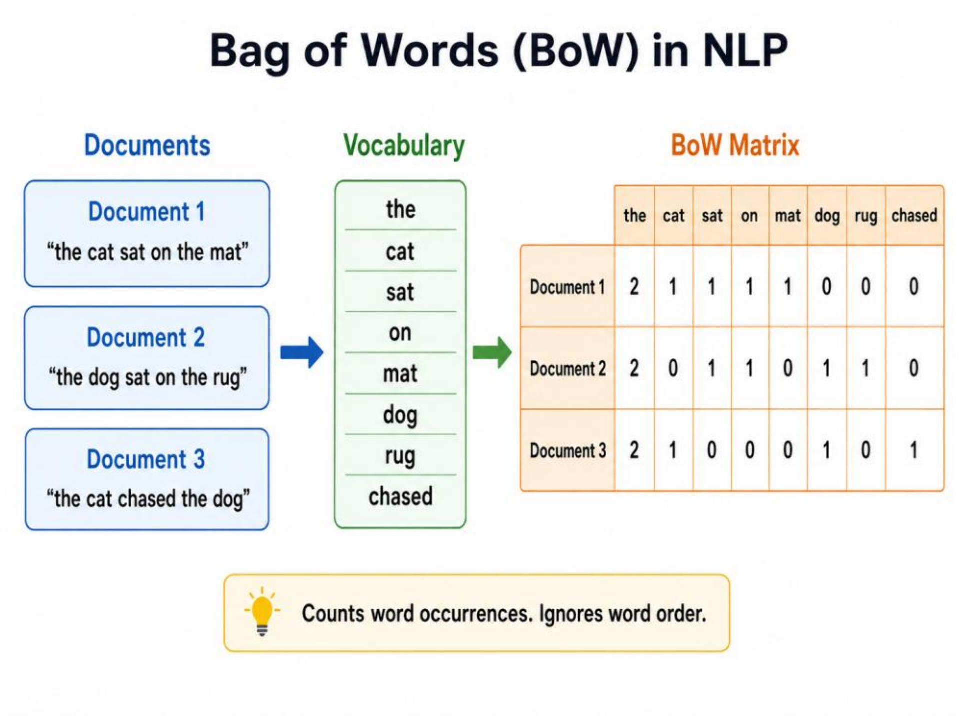
FROM TEXT TO NUMBERS

- Machine learning models need numeric input.
- Text representation is the bridge between language and models.
- Different representations preserve different kinds of information:
    - frequency,
    - importance,
    - similarity,
    - sequence,
    - context.



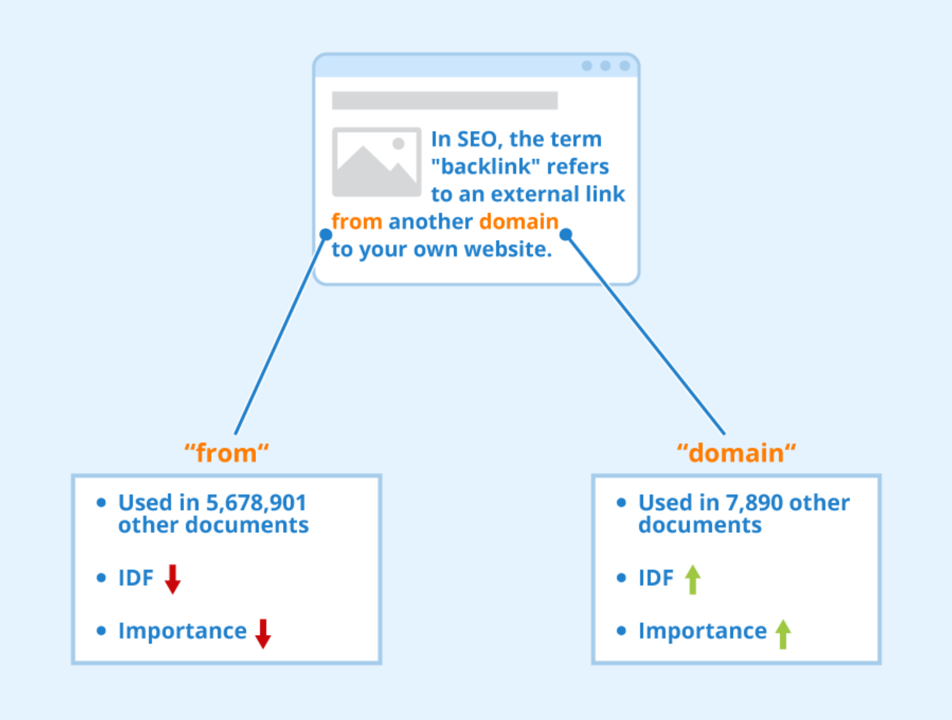
BAG OF WORDS

- Represents a document by word counts.
- Ignores word order.
- Simple, interpretable, and useful for many baseline models.
- Works well when keywords strongly indicate categories or sentiment.
- **Limitation:** loses grammar and context.



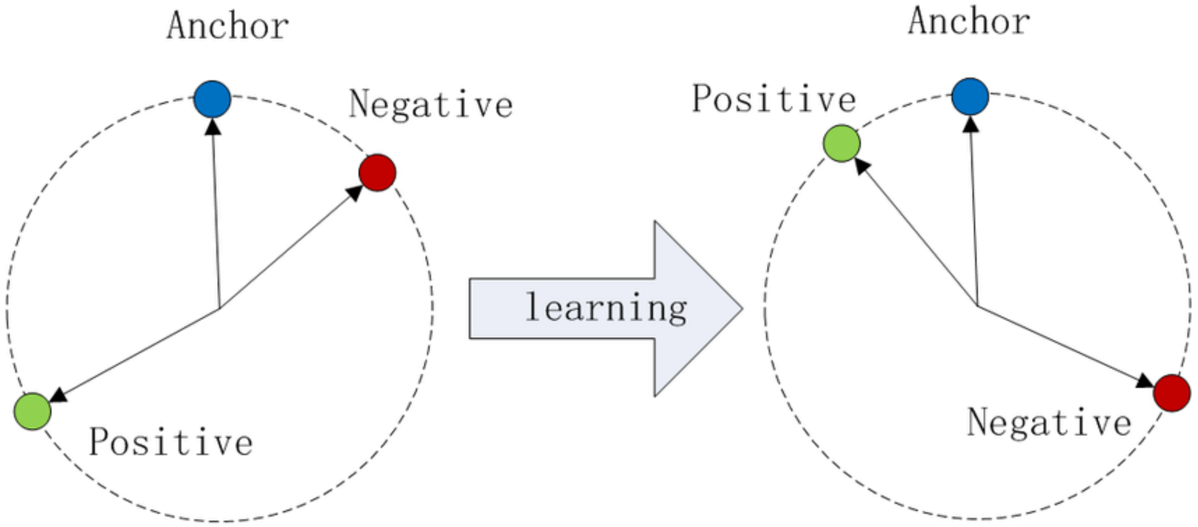
TF-IDF

- Term Frequency–Inverse Document Frequency (TF-IDF) weights words by how important they are to a document.
- Frequent words in one document get higher weight.
- Very common words across many documents get lower weight.
- Useful for search, document similarity, classification, and retrieval.



COSINE SIMILARITY

- Documents can be compared as vectors.
- Cosine similarity measures direction, not raw length.
- Useful for finding similar documents, recommendations, search, and clustering.
- Works with BoW, TF-IDF, and embedding-based vectors.



LIMITATIONS OF FREQUENCY-BASED TEXT MODELS

“Car” and “automobile” may look unrelated.

Word order is lost.

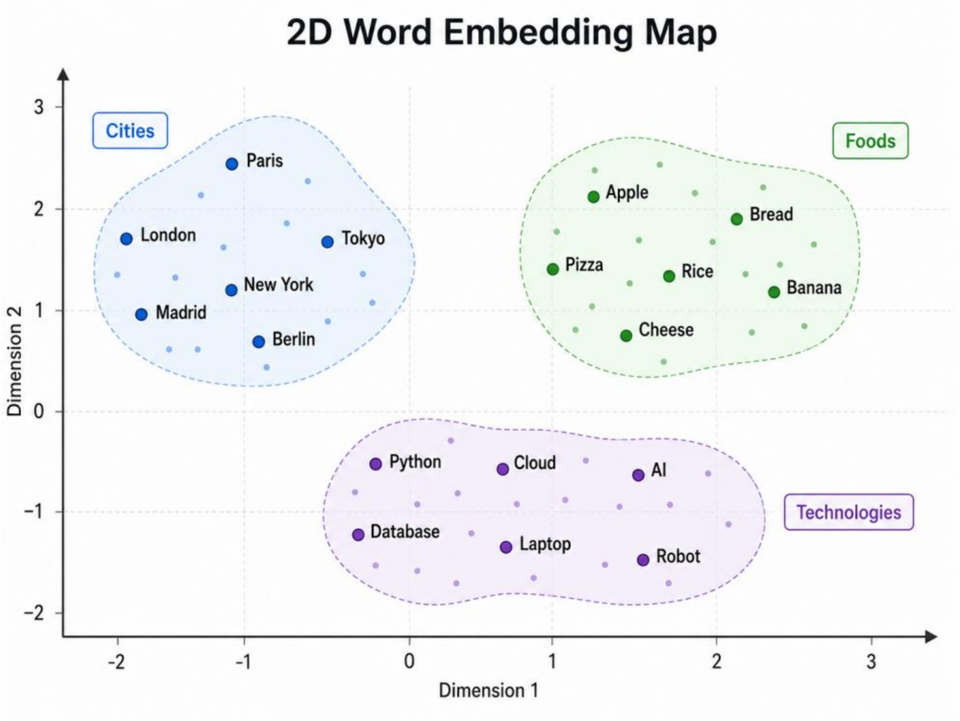
Negation can be missed: “not good” vs. “good”.

Context and meaning are only weakly represented.

This motivates embeddings and language models.

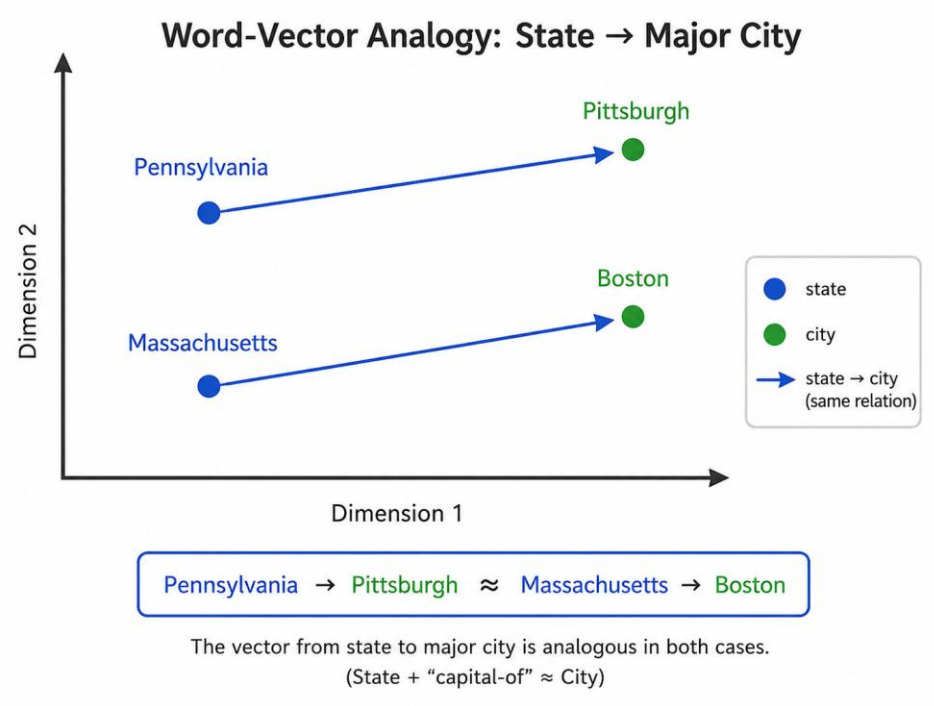
WORD EMBEDDINGS

- Embeddings represent words as dense vectors.
- Similar words appear closer in vector space.
- They capture semantic relationships better than one-hot or count vectors.
- Word2Vec and GloVe are classic examples.
- Embeddings can be reused from pre-trained models.



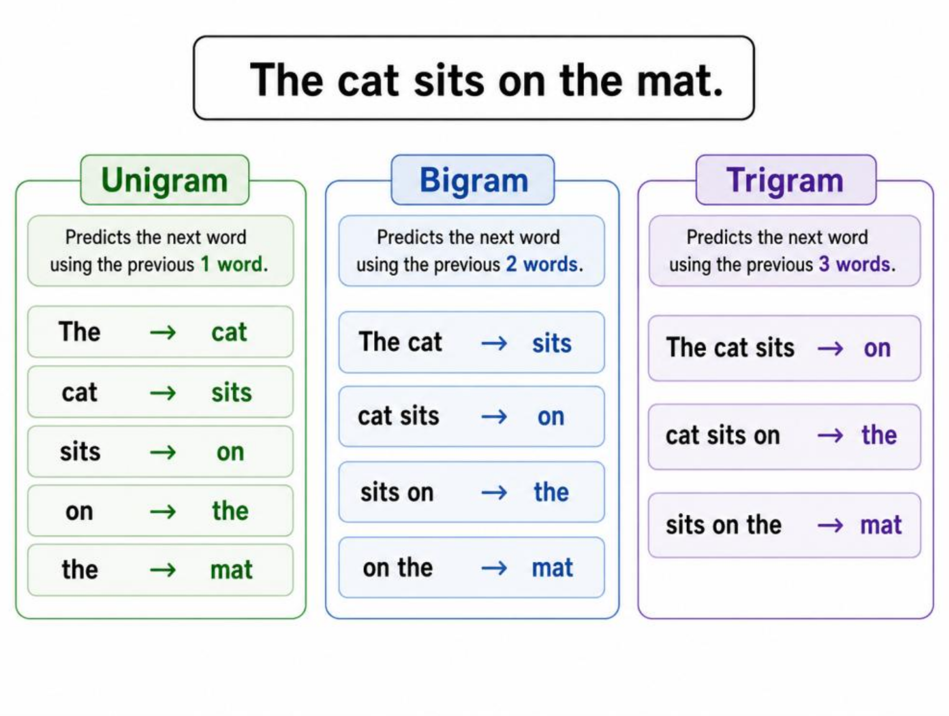
WORD VECTORS AND ANALOGY

- Embeddings can encode relationships.
- Classic idea: vector arithmetic can approximate analogies.
- Example: country → capital, gender relations, category relations.
- These properties are useful but not perfect or neutral.



LANGUAGE MODELS AND N-GRAMS

- Language models estimate the probability of word sequences.
- N-gram models predict the next word using the previous N-1 words.
- They capture local context.
- Useful for text generation, author identification, autocomplete, and comparison.
- Limitation: they struggle with long-range meaning.

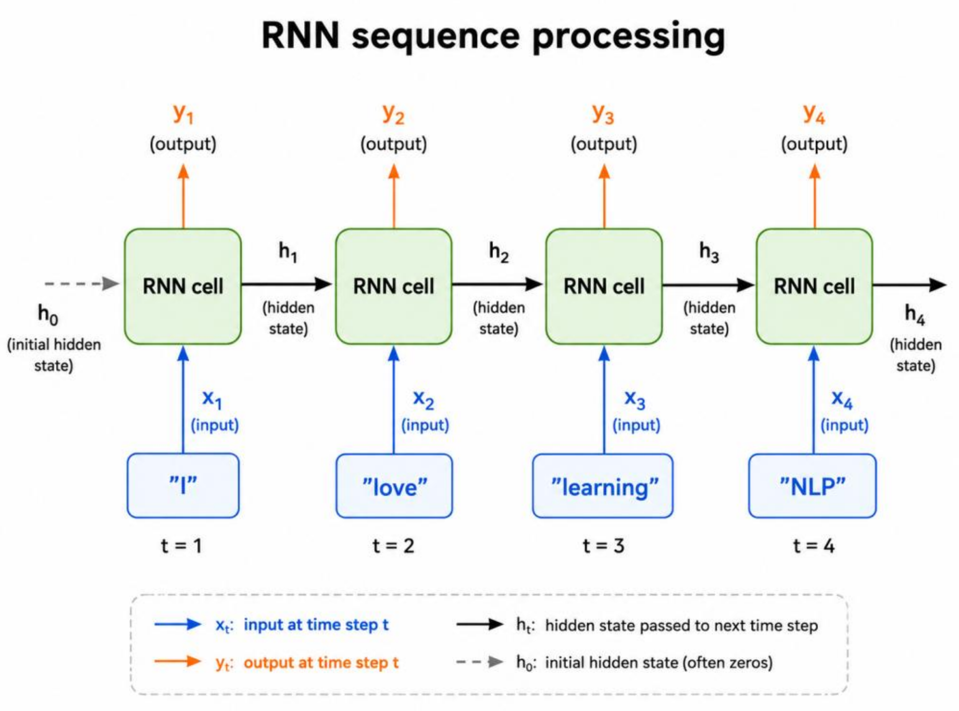


EVALUATING LANGUAGE MODELS

- Accuracy
- Precision
- Recall
- F1
- BLEU
- ROUGE
- Perplexity

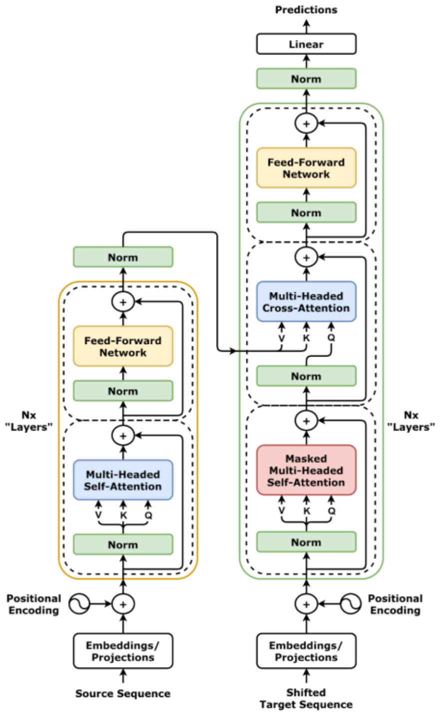
ADVANCED NLP: RNNS, LSTM, AND GRU

- Recurrent Neural Networks (RNNs) process sequences step by step.
- They keep a form of memory from previous tokens.
- Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) improve long-sequence learning.
- Historically important for translation, language modeling, and sentiment analysis.
- **Limitation:** harder to parallelize and weaker on very long contexts.



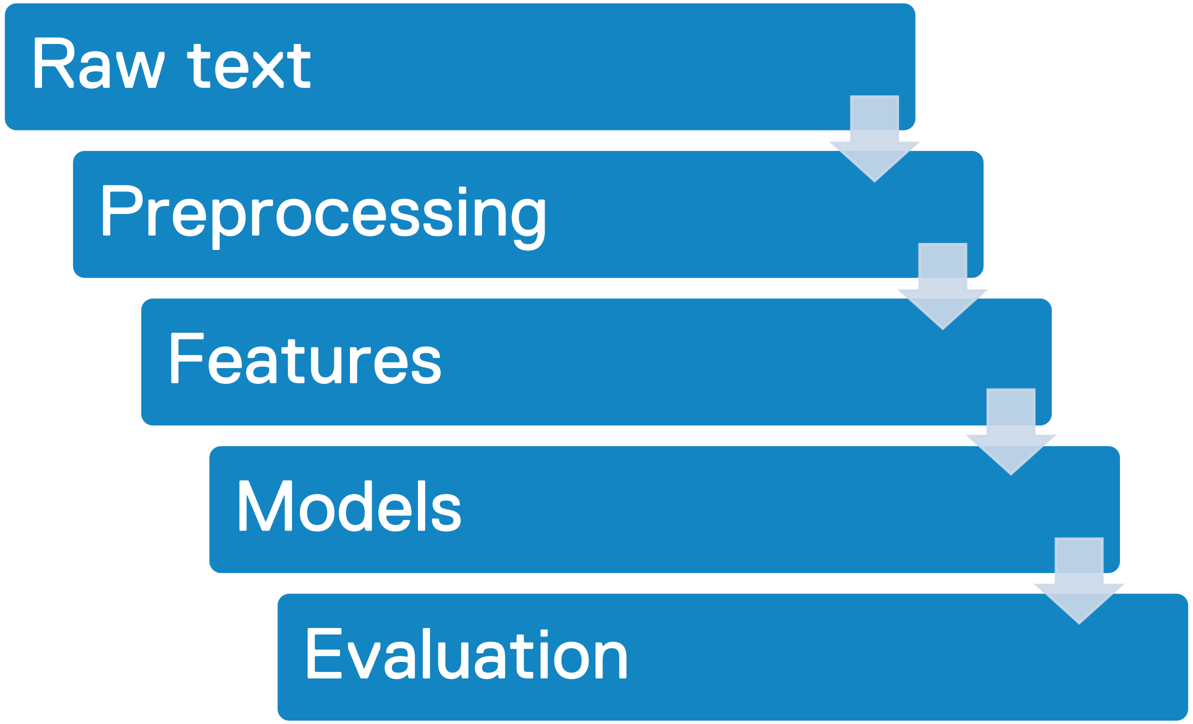
TRANSFORMERS AND TRANSFER LEARNING

- Transformers use attention to focus on relevant parts of the text.
- They process sequences more efficiently than classic RNNs.
- BERT-style models are strong for understanding tasks.
- GPT-style models are strong for generation tasks.
- Transfer learning adapts pre-trained models to specific tasks.



NLP PROJECT PIPELINE AND FINAL RECAP

- Start with the task: classification, generation, search, extraction, summarization, or labeling.
- Collect and inspect text ethically.
- Preprocess based on task needs.
- Choose representation and model.
- Evaluate with suitable metrics: accuracy, precision, recall, F1, BLEU, ROUGE, perplexity.
- Remember: NLP is not only modeling; it is data, representation, evaluation, and responsible use.



Technology Elevated